CS 412 Intermediate Project Report
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1. **Description of the data (Yash)**

For our project we will be using the Cassava Leaf Disease Classification dataset from Kaggle as this dataset has/contains 21,367 images of cassava leaves that were collected from the fields in Uganda and each of the image is labeled as either a healthy leaf or one of the four different cassava diseases. Our goal is to use these images to train and compare the different machine learning models that can identify the disease from a picture of a cassava leaf. One of the main challenges with this dataset is that the images are not equally distributed among all the five classes and for example, Cassava Mosaic Disease (CMD) makes up approximately 61 percent of the dataset and while the rarest disease class only makes up about 5 percent of the dataset. This creates a problem because a machine learning model will become better at predicting the majority class while having difficulty identifying the diseases that have fewer/lesser images. This is important for our project because we are not only trying to achieve/get high overall accuracy but we also want our models to be able to identify the rare diseases correctly as to address this problem we plan to compare the different approaches for handling the imbalanced in the dataset, focal loss, including oversampling, and class balanced loss. We will use the images as a input for our machine learning models and then comparing their performance using accuracy, recall for each disease, confusion matrices, and macro f1 score as this will help us understand how well each model performs across all the five classes rather than only focusing on the overall accuracy.

4. What remains to be done (Yash)

For the remaining parts of our project we will be focusing on completing our model comparisons and testing the different methods to improve the identification of the rare cassava leaf diseases. We will first be making sure that our classical baseline model is using HOG and SVM as well as/along with our two CNN models, EfficientNet-B0, and ResNet-50 have been trained and have been evaluated so that we can compare their performances. After comparing these models we will focus on the handling of the class imbalance in the dataset as we plan to test the three different approaches that are balanced loss, oversampling, and focal loss. Oversampling will allow the model to see images from the rare disease classes more frequently during the training. Class balanced loss will give more importance to classes with the fewer images and lastly, while focal loss will help the model focus more on the examples that are difficult to classify. Once we have completed all these experiments we will compare the results using confusion matrices, macro F1 score, accuracy, and recall for each disease as this will help us see which approach performs better and especially/particularly for the rare diseases. According to our project schedule we plan to complete these experiments between 2nd November and November 14th. After that we will be working together on preparing our final presentation, that will be including a possible demonstration use a cassava leaf image and lastly, finally we will organize our findings, discuss the results among ourselves, and complete our final project report by December 1st 2026.